

LegalEase: AI-Powered Legal Document Generator

Project Testing Phase

1. Team Details

Team Name: LegalEase: AI-Powered Legal Document generator

Team Members:

- 1. Bharathkumar A
- 2. Gurusankar S
- 3. Bikash Thaabaa R
- 4. Sabarivasan S

2. Introduction

- The Project Testing phase verifies that all major features of LegalEase work correctly according to the requirements and design.
- Testing helps identify errors, improve reliability, and ensure that users can generate, edit, and download legal document drafts successfully.

3. Testing Objectives

- Verify that the application works according to the specified requirements.
- Check communication between the Streamlit frontend and FastAPI backend.
- Verify integration with the Gemini AI service.
- Check generated document content and preview.
- Verify editing functionality.
- Test PDF and Word document downloads.
- Identify and correct errors before final submission.

4. Types of Testing

- Unit Testing – individual functions and modules are checked.
- Integration Testing – frontend, backend, AI module, and document export are tested together.
- Functional Testing – application features are checked against requirements.
- Input Validation Testing – valid, incomplete, and incorrect inputs are tested.
- UI Testing – Streamlit interface elements and user interactions are checked.
- Output Testing – generated documents and downloaded files are checked.

5. Test Environment

- Operating System: Windows
- Programming Language: Python
- Frontend: Streamlit
- Backend: FastAPI with Uvicorn
- AI Service: Google Gemini
- Development Tool: Visual Studio Code
- Version Control: GitHub

6. Test Cases

Test Case	Test Description	Expected Result
TC01	Open the Streamlit application.	Application should open successfully.
TC02	Select a document type.	Selected document type should be accepted.
TC03	Enter parties, terms, and effective date.	Input values should be accepted correctly.
TC04	Click Generate Document.	Request should be sent to the FastAPI backend.
TC05	Generate a document using Gemini.	A structured document draft should be returned.
TC06	Display generated document.	Generated content should appear in the preview.
TC07	Edit generated content.	User should be able to modify the document.
TC08	Download as Word.	A valid Word document should be downloaded.
TC09	Download as PDF.	A valid PDF document should be downloaded.
TC10	Test incomplete required input.	Application should handle missing input appropriately.
TC11	Test API/server error.	Application should handle the error without crashing.

7. Testing the AI Document Generation

The AI generation feature is tested using sample inputs such as NDA, Employment Contract, and Lease Agreement. The output is checked to ensure that the document type, parties, terms, and effective date are reflected in the generated draft.

8. API Testing

The FastAPI /generate endpoint is tested with valid request data. The response is checked to confirm that the backend returns the generated document. Error cases are also tested.

9. Document Export Testing

The PDF and Word download features are tested after generation and editing. Downloaded files are opened to verify that the content is present, readable, and properly formatted.

10. Error Handling

Common issues such as missing input, API communication errors, AI service errors, dependency problems, and document formatting issues are checked and corrected where required.

11. Testing Outcome

Testing verifies the major LegalEase workflow from user input through AI generation, preview, editing, and document export. Issues identified during development are corrected to improve reliability.

12. Project Testing Conclusion

The Project Testing phase ensures that LegalEase meets its planned functional requirements and that the main modules work together correctly. Successful testing supports the project's readiness for demonstration and final submission, subject to review of generated legal content.